

Matsuyama in Space

Agriculture, connectivity and Korea's escape from poverty, 1965–85

Where the model stands after the fresh look

Korea growth project — working slides for coauthors

4 October 2026

Roadmap

1. Narrative
2. Core mechanisms
3. Theory: the current model
4. Calibration and data

One-line summary

We re-centred the project on the *macro* story. The baseline is now Korea's 1965–85 history, and the core mechanism is a spatial version of Matsuyama (1992): whether farm productivity growth **retains** or **releases** labour depends on each township's access to crop markets and to non-farm jobs.

Narrative

The question

How did Korea escape poverty between 1960 and 1990, and through which mechanisms?

Candidates: industrial parks, roads, rural transformation, world demand, capital, schooling.

Margin	~1960/65	~1980/90
Real GDP per capita	1	×4 (1985) [†]
Agriculture, employment share	59% (1965) [†]	25% (1985) [†]
Manufacturing, VA share	15% (1965)	30% (1985)
Urban population share	28% (1960)	57% (1980), 74% (1990)
Farm household population	~14m	<6m (1990)
Exports / GDP	~9% (1965) [†]	~33% (1985) [†]
Absolute poverty	~2/5 (mid-60s) [†]	~1/10 (1980) [†]

Unmarked: paper draft / WDI. [†] order-of-magnitude values, not yet checked against source (*scripts/data_targets.py*).

- One of the fastest structural transformations on record.
- Poverty was overwhelmingly **rural**: the countryside is where parks, roads and rural programmes should matter most.
- The township panel being digitised (1960–90, annual) lets us see the transformation *where it happened*.

From “perturb a stationary economy” to “subtract from history”

Old design

Calibrate a 1965 economy, switch on HCI, report policy — baseline.

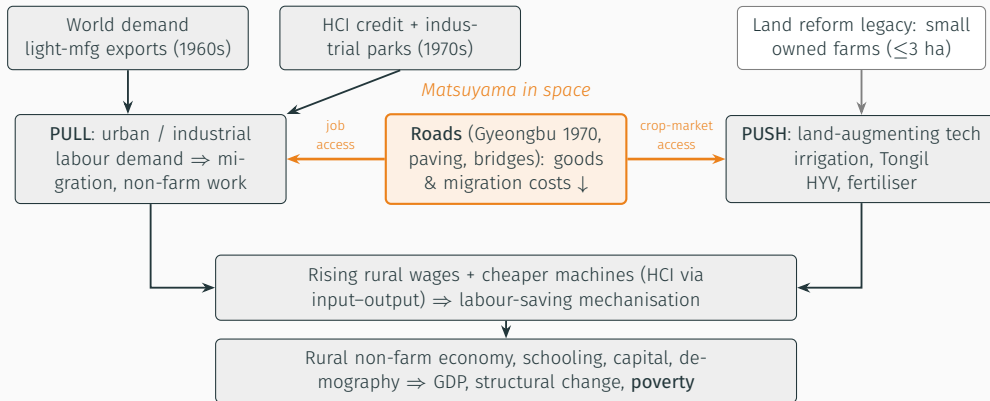
What the fresh look found when the code was run against the macro story:

- Trade closure was **indeterminate**: wages differing by 65% all solved to 10^{-12} .
- 1965 economy not Korea: exports 73% of GDP, farm employment 11%.
- **No growth engine**: baseline GDP $\times 1.12$ over 1965–85 (data $\times 4$); HCI supplied $\sim 2/3$ of all model growth.
- Mechanisation and export margins un-nested; agriculture at a corner.

New design: baseline = history

- The baseline reproduces the observed transformation.
- Programmes enter as *measured* shocks with *identified* elasticities.
- Counterfactuals **subtract** a shock from history; outcomes are decomposed (Shapley), not differenced.
- Township designs identify *relative* effects; the model supplies the general-equilibrium **intercept** (Nakamura–Steinsson; Adão–Arkolakis–Esposito).
- Headline outcomes: poverty and structural transformation, GDP alongside.

The mechanism chain



Each arrow needs a model block, a **measured** shock, an **identified** elasticity, and a counterfactual that switches it off.

Matsuyama (1992), and why Korea makes it spatial

Matsuyama (1992, JET): manufacturing has learning-by-doing.

- **Closed economy.** Farm productivity $\uparrow \Rightarrow$ food needs met with fewer workers (Engel) $\Rightarrow$ labour **released** to industry $\Rightarrow$ faster growth. The “food problem” (Gollin–Parente–Rogerson 2007).
- **Open economy.** Farm productivity $\uparrow \Rightarrow$ comparative advantage in farming $\Rightarrow$ labour **retained** $\Rightarrow$ slower industrialisation.

The Korean narrative implicitly assumes the closed case. Which case applies is usually a national question.

Why Korea

After 1969 the rice market was **nationally closed and policy-priced** (import controls, government procurement under the dual grain-price policy).

A township was *open to the nation* while the nation was *closed to the world*.

So the “regime” is township-level by construction, and roads move townships between regimes.

The hypothesis, stated carefully

Hypothesis

Agricultural productivity growth induces **commercialisation** where access to crop demand dominates, and **labour release** where access to non-farm opportunities dominates.

- A connectivity-dependent *sign reversal* of the farm-employment response is one derivable implication, **not** the test.
- The test: the model, with trade, distance and Engel parameters estimated elsewhere, predicts an interaction coefficient X ; the reservoir event studies give Y . A **held-out moment**.
- Related: Gollin–Rogerson (2014), Sotelo (2020), Foster–Rosenzweig (2004), Bustos–Caprettini–Ponticelli (2016).

Novelty

Cheung–Yang (2024, SSRN 4768158; reportedly JIE, to verify): Indian roads lower machine prices relative to wages and induce labour-saving technology. So roads → mechanisation alone will not distinguish us. Our contribution: separately identify *farm productivity shocks*, *industrial labour-demand shocks* and *connectivity*, and show where their interaction yields commercialisation versus release.

Core mechanisms

A township in one equation

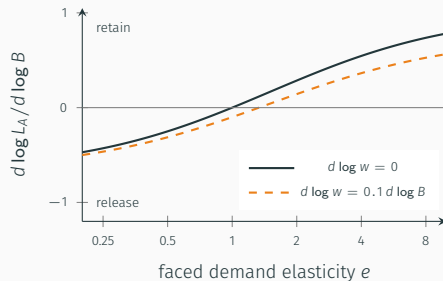
Competitive farms, $Q_i = (B_i T_i)^\alpha L_{Ai}^{1-\alpha}$; land-augmenting shock B_i (reservoir). Local crop demand faced with elasticity e_i ; opportunity wage w_i .

$$d \log L_{Ai} = \frac{\overbrace{\alpha (e_i - 1) d \log B_i}^{\text{commercialisation vs. food problem}} - \overbrace{e_i d \log w_i}^{\text{labour release}}}{\alpha e_i + 1 - \alpha}$$

- From the FOC

$$\log L_A = \frac{1}{\alpha} [\log p + \alpha \log BT - \log w] + c \text{ and } d \log p = -\frac{1}{e} d \log Q.$$

- Goods-market access** raises e_i : isolated $\approx$ local food elasticity (< 1); connected $\rightarrow \theta$ (4-8).
- Labour-market access** moves w_i (industrial pull, migration).
- Fixed w : labour retained iff $e_i > 1$.



Illustrative, $\alpha = 0.4$. Not the quantitative model.

Why the sharp version (“the sign should flip”) is not the test

1. **Gravity builds in the answer.** Under Armington / EK a connected township faces $e = \theta > 1$, an isolated one the food elasticity < 1 . The flip holds by construction. The data identify **where** on the access distribution the crossing occurs, which disciplines distance costs and Engel parameters. (Matsuyama’s open case also needs a homogeneous traded good; Armington removes it.)
2. **Connectivity moves both terms.** Connected townships also have non-farm jobs, cheaper machines, lower migration costs: $d \log w > 0$ pushes towards release, opposite to comparative-advantage retention. A null interaction is uninterpretable without separate industrial-pull controls.
3. **Irrigation is not Hicks-neutral.** Korean paddy irrigation + Tongil HYV raised labour per hectare (double cropping, transplanting). A labour-using shock retains labour everywhere. Estimate the factor bias jointly.

Two reconciliations

(i) **Rice-price support** from 1969 insulates farm-gate prices from local supply, weakening the isolated-township price channel: derive price policy and prediction together. (ii) **Township retention $\neq$ aggregate retardation**: the growth conclusion needs manufacturing learning-by-doing.

Labour-saving technology: mechanisation with an indivisible machine

A tiller cuts labour per hectare by a fraction χ and costs $u_t = p_{m,t}(r_t + d)$ a year. A farm of size ℓ adopts iff

$$\chi \lambda_0 \ell w_{iA,t} \geq u_t \iff \ell \geq \ell_{it}^* = \frac{p_{m,t}(r_t + d)}{\chi \lambda_0 w_{iA,t}}, \quad \text{adoption}_{it} = 1 - F_i(\ell_{it}^*).$$

- F_i = **observed** landholding distribution (land-reform legacy, 3-ha ceiling); custom hiring smooths the threshold.
- Adoption responds to local farm wages (**industrial pull**), machine prices (**HCI machinery** via input-output) and farm size.
- Hayami–Ruttan induced innovation with a micro-founded extensive margin.

The distinction that matters

Labour-saving change **releases** labour; land-augmenting change can **absorb** it (Bustos–Caprettini–Ponticelli 2016).

In the current code this margin is a reduced form: a fixed-cost mechanised technology with $\xi = 1.2$ (next section).

The agricultural productivity gap needs sectoral wages

With one wage per location, farm and non-farm value added per worker differ only through labour shares λ^k :

$$\text{APG}_d \equiv \frac{\text{VA}_d^A / L_d^A}{\text{VA}_d^N / L_d^N} = \frac{w_d^A / \lambda_d^A}{w_d^N / \lambda_d^N} \stackrel{w^A \equiv w^N}{=} \frac{\lambda_d^N}{\lambda_d^A} \gtrsim 1.$$

- Land takes a large share of farm VA, so an integrated market delivers APG near or above one. The data (0.38 VA / 0.59 employment in 1965[†]) imply ≈ 0.43 .
- In the calibrated baseline, the 1965 farm VA share is 0.595 against 0.38[†] in the data: the integrated market cannot open the gap.
- Fix: **occupation choice** within location, with a non-pecuniary occupation value b (attachment to farming, barriers to leaving it). Gollin–Lagakos–Waugh (2014); Hsieh–Hurst–Jones–Klenow (2019).
- Policy content: if roads and schooling lower the farm wedge, that is a mechanism of structural change in its own right, and the decomposition prices it.

Theory: the current model

Sequence of static spatial equilibria, state = L_{t-1} .

- Locations $d \in \mathcal{N}$ (5 stylised: Seoul, Busan, Changwon, Daegu, Rural); periods 1965/68/73/75/80/85.
- **Sectors:**
 - **Rice:** protected, internationally non-traded; carries the mechanisation margin.
 - **OtherAg:** import-competing at world prices; not exported.
 - **Manufacturing:** traded; the export sector.
 - **Services:** non-traded.
- Input-output linkages; sector-specific agglomeration $L_{o,t-1}^{\rho_j}$.
- PIGL non-homothetic demand (Boppart 2014).
- Gumbel migration (myopic), now **nested** with occupation choice.
- Monopolistic competition with bounded Pareto draws; **nested** entry / adoption / export choice.
- Local land ownership; nationally pooled profits; government buys output.
- Small open economy; numeraire: foreign manufacturing price; net exports **pinned exogenously**.
- Growth engine: sectoral TFP growth + rising foreign demand for manufactures.

Preferences: PIGL and the food problem

Indirect utility and expenditure shares at income y , prices P_{dj} , $\mathcal{P}_d = \prod_j P_{dj}^{\alpha_j}$:

$$V_d = \frac{1}{\eta} \left(\frac{y}{\mathcal{P}_d} \right)^\eta - \sum_j v_j \log P_{dj}, \quad \psi_{dj} = \alpha_j + v_j \left(\frac{y}{\mathcal{P}_d} \right)^{-\eta}, \quad \sum_j \alpha_j = 1, \sum_j v_j = 0.$$

- $\eta > 0$, $v_{\text{food}} > 0$: the food share falls with real income (Engel), services rise. Calibrated $\eta = 0.39$.
- Shares checked in $[0, 1]$ along the solved path, not just imposed.
- Exact aggregation over finitely many income groups, so farm/non-farm income differences change the *composition* of local demand.
- Migration and welfare use the **exact equivalent income** $\mathcal{Y} = \mathcal{P}_0[\eta(V + z_0)]^{1/\eta}$, a monotone transform of V for every η . (The old index xe^{-z} was not, and had the wrong income elasticity.)
- Limitation: Cobb–Douglas price aggregator, so no Baumol relative-price channel. Non-homothetic CES (Comin–Lashkari–Mestieri) if the data call for it.

Location × occupation choice

Agent in origin o chooses location d and occupation $\kappa \in \{\text{farm, mfg, services}\}$ to maximise $W_{od} + \log b_{d\kappa} + \log \mathcal{Y}_{d\kappa} + v_{i,d\kappa}$, with GEV shocks $G(x) = \sum_d (\sum_{\kappa} x_{d\kappa}^{\varepsilon})^{\nu/\varepsilon}$:

$$\pi_{\kappa|d} = \frac{(b_{d\kappa} \mathcal{Y}_{d\kappa})^{\varepsilon}}{\sum_{\kappa'} (b_{d\kappa'} \mathcal{Y}_{d\kappa'})^{\varepsilon}}, \quad \Phi_d = \left[\sum_{\kappa} (b_{d\kappa} \mathcal{Y}_{d\kappa})^{\varepsilon} \right]^{1/\varepsilon}, \quad \mu_{od} \propto \left(\frac{\bar{V}_d L_{d,t-1}^{\nu} \Phi_d}{\delta_{od}} \right)^{\nu}.$$

- Random-utility consistent iff $\nu \leq \varepsilon$: switching sector within a location is easier than moving.
- Each location–occupation labour market clears at its own wage
 $w_{d\kappa} L_{d\kappa} = \sum_{j:o(j)=\kappa} [\text{labour share of } TFC_{dj} + \text{fixed-cost bill}]$.
- Rice and other crops are *one* occupation: they are products, not labour markets (two would double farming's taste mass).
- Welfare: $\partial \mathbb{E}U_o / \partial \log \mathcal{Y}_{d\kappa} = \mu_{od} \pi_{\kappa|d}$ (tested).
- Integrated limit: $\varepsilon \rightarrow \infty$, $b = 1$ gives one wage per location.

What the occupation block buys

Wage-gap decomposition (exact in equivalent incomes)

$$\log \frac{y_{dA}}{y_{dN}} = \underbrace{\frac{1}{\varepsilon} \log \frac{L_{dA}}{L_{dN}}}_{\text{supply term}} - \underbrace{\log \frac{b_{dA}}{b_{dN}}}_{\text{non-pecuniary wedge}}$$

1. $b = 1$ is **not frictionless**. It means equal pay $\Rightarrow$ equal occupation shares. A 60% farm share at a *lower* farm wage needs $b_A > b_N$.
2. **The gap moves with structural change**. When farm labour demand falls (Engel, import competition, mechanisation), finite ε keeps workers in farming at a falling relative wage: the gap **widens** before switching and migration catch up.
3. **Sectoral labour supply**: $\partial \log L_{d\kappa} / \partial \log y_{d\kappa} = \varepsilon(1 - \pi_{\kappa|d}) e_{d\kappa}^y$.
4. **Taste vs. selection**. A Roy/Fréchet version (Lagakos–Waugh 2013) is a tractable extension, and predicts different earnings changes for switchers (Hamory et al. 2021).

Production: nested entry, adoption and export

With $x = \phi^{\sigma-1}$, every option's profit is linear in x : option k has slope a_k and fixed cost c_k . Agriculture: exit $(0, 0)$, traditional (a^T, F) , mechanised $(a^M, F + \check{F})$, mechanised exporter $(a^M + a^X, F + \check{F} + \tilde{F})$. The firm picks the upper envelope:

$$\text{"option } k \text{ or higher"} \iff x \geq X_k, \quad X_k = \min_{k' \geq k} \max_{i < k'} \frac{c_{k'} - c_i}{a_{k'} - a_i}.$$

Ordering $\bar{\phi} \leq \check{\phi} \leq \tilde{\phi}$ holds by construction; adopters and exporters are active firms. A traditional segment exists iff

$$\frac{\check{F}}{F} > \Delta, \quad \frac{\check{F} + \tilde{F}}{F} > \Delta + \frac{a^X}{a^T}, \quad \Delta \equiv \left(\xi \frac{c}{\check{c}} \right)^{\sigma-1} - 1.$$

- Δ rises with w/P_{mfg} : the mechanised technology substitutes machinery for labour, so adoption turns on as **wages rise and machines get cheaper** (induced innovation). Hence $\xi = 1.2 \approx$ the 1965–85 range of $\check{c}/c$; fixed costs are in labour units.
- The old pairwise cutoffs had $\tilde{\phi} < \check{\phi} < \bar{\phi}$: farms exported and mechanised without entering.

External balance pins the wage level

The old closure set the foreign transfer

$T = \text{IM} - \text{EX}$ at the current guess. Household budgets + goods-market clearing already imply this at any fixed point, so one equation was lost.

One-good example: $d(w)E + X(w) = wL$ and $E = wL + T$ with $T = (1 - d)E - X$ are the same equation $\Rightarrow$ every w solves.

Fix: $\text{EX} - \text{IM} = \text{NX}^*$ exogenous (default 0; later the observed path, deficits financing investment). The domestic price level clears the foreign account. Guess-invariant to 3×10^{-13} .

guess $\times k$	0.5	1.0	2.0
w_{Seoul}	0.291	0.367	0.480
$(\text{EX} - \text{IM})/\text{abs.}$	-0.37	-0.56	-0.67
mean real income	0.96	1.92	3.50

Old closure, residuals $< 10^{-11}$ in every column.

Consequence: the aggregate export elasticity σ_x (separate from σ) now does real work. Old 3-sector HCL headline, 1985 heavy-mfg gross-output share effect:

σ_x	2	4	6
effect	+0.021	+0.103	+0.137

Equilibrium and solution

Given L_{t-1} and period- t fundamentals, an equilibrium is $\{w_{d\kappa}, r_d, P_{dj}, E_{dj}, L_d, \bar{\tau}, \bar{\pi}\}$ such that:

1. **Migration and occupation:** $\mu_{od}, \pi_{\kappa|d}$ as above; $L_d = \sum_o \mu_{od} L_{o,t-1}$, $L_{d\kappa} = \pi_{\kappa|d} L_d$.
2. **Final demand:** PIGL shares summed over occupation income groups.
3. **Firms:** nested cutoffs, productivity aggregates, price indices, revenues.
4. **Expenditure:** $E = E^F + E^M + E^G$; intermediates from true factor cost $TFC = \frac{\sigma-1}{\sigma(1-s)} Y$.
5. **Factor markets:** each location–occupation labour market and each land market clears; fixed costs are paid to local labour.
6. **Government:** $\bar{\tau}$ finances subsidies and infrastructure (which buys goods); $\bar{\pi}$ rebates pooled profits (sign-unrestricted).
7. **External balance:** $EX - IM = NX^*$.

Damped fixed point in $(w, r, P, E, \bar{\tau}, \bar{\pi})$; relative occupation wages by a log-Newton step $\Delta \log w_{d\kappa} = \log(D_{d\kappa}/S_{d\kappa})/(1 + \varepsilon)$. Walras, resource and budget identities hold to $\sim 10^{-13}$ (tested).

Amenity inversion: with L_{1965} pinned, the rest of the equilibrium does not depend on $\bar{V}$, so $\bar{V}$ is recovered exactly from the logit. The observed 1965 population is an equilibrium; later urbanisation is a **model outcome**.

Where the model is heading: five principles

Principle	Content	Replaces
1. Baseline = history	Annual 1962–90; invert fundamentals each year (once identified); programmes as measured shocks; Shapley decomposition	Hand-set paths; 6 uneven periods; policy — baseline
2. Simplify production	CES gravity (Armington/EK) with sectoral external economies in employment density	Melitz everywhere
3. People in sectors	Nested location × occupation incl. non-employment (Lewis surplus labour)	Single regional wage
4. Agriculture on observed land	Competitive farm households on paddy and dry fields; farm-size classes; irrigation, HYV; mechanisation threshold in w/p_m	Monopolistic “farms” with fixed adoption cost
5. Growth engines	Capital with HCI credit wedges and foreign saving; schooling; demography; world demand; poverty and welfare as outputs	Output subsidies; fixed population

$$\pi_{in,k} = \frac{(\tau_{in,k} c_{ik} / A_{ik})^{-\theta_k}}{\sum_l (\tau_{ln,k} c_{lk} / A_{lk})^{-\theta_k} + (\tilde{\tau}_{n,k} \tilde{p}_k)^{-\theta_k}}, \quad A_{ik} = \bar{A}_{ik} \left(\frac{L_{ik}}{\text{area}_i} \right)^{\rho_k}, \quad \tau_{in,k} = e^{\varphi_k \text{time}_{in}} a_i a_n.$$

Done so far: (3) without non-employment; (4) only as Rice/OtherAg split; (5) only TFP + world demand.

The annual inversion is not identified as written

Per township-year under principles 2–3:

Unknowns	K productivities + $(K - 1)$ wedges + 1 amenity	$= 2K$
Observables	population + $(K - 1)$ employment shares	$= K$

In $\pi_{k|i} \propto (b_{ik}w_{ik})^\varepsilon$, higher A_{ik} raises w_{ik} and hence $\pi_{k|i}$ exactly as higher b_{ik} does.

- Need wages, output, prices or trade flows: county MMS wages and VA, provincial farm wages, crop output and prices. Or restrictions: wedges common within county/province, or constant up to trend.
- Demonstrate identification (analytically or by recovery on synthetic data) **before** building ***inversion.py*** beyond amenities.
- Inverted fundamentals are not causal policy effects: event studies on them need the same assignment design and must carry inversion uncertainty.
- Exact fit is not validation: inversion inputs cannot also validate.

Calibration and data

History calibration: 8 parameters, 8 targets

Parameter	Value	Main target
g_{mfg} (TFP growth)	4.1%/yr	real GDP p.c. 1985/65
$g_{\text{farm}} = g_{\text{svc}}$	3.2%/yr	mfg VA share 1985
η (PIGL)	0.39	farm emp. share 1985
v_{food} (Engel shifter)	0.51	farm emp. share 1965
mfg share of non-food α	0.02 (bound)	mfg VA share 1965
foreign demand, 1965 level	0.0044	exports/GDP 1965
foreign demand growth	24%/yr	exports/GDP 1985
urban non-farm TFP premium	1.17	urban share 1980

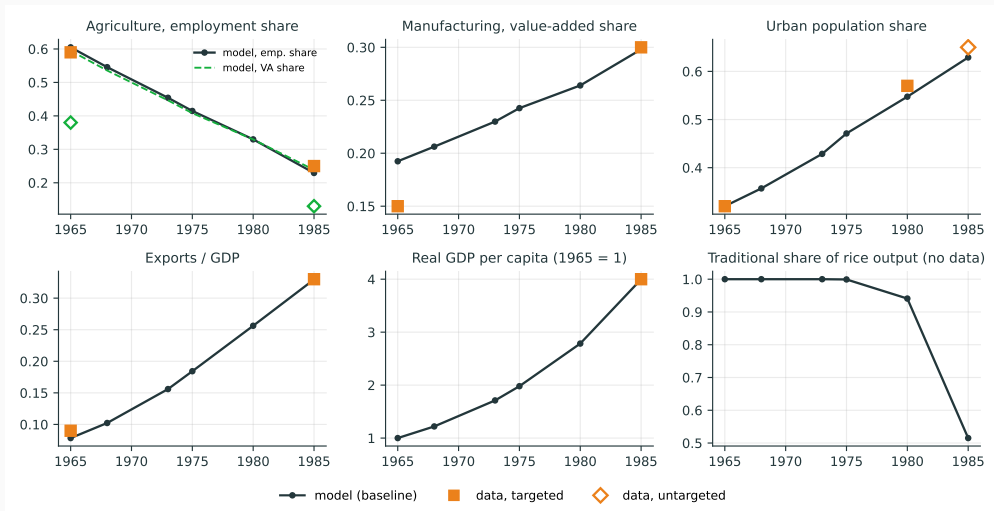
scripts/calibrate_history.py: least squares; each evaluation first inverts 1965 amenities; PIGL share bounds as a penalty.

Held fixed (from earlier calibration or assumption): $\sigma = 4$, $\theta = 5$, $\nu = 5$, $\xi = 1.2$, ρ_j , IO coefficients, land shares, trade and migration cost matrices.

Assumed, no data yet:

- rice = half of food demand
- urban farm TFP = $0.4 \times$ rural
- food asymptote $\alpha_{\text{food}} = 0.05$
- 1965 city split stylised
- rice mechanisation fixed cost
- services TFP growth = farm

The baseline reproduces the 1965–85 transformation



Solved at the current calibration (*make_figures.py*). Green: farm value-added share (model, dashed; data, hollow).

Fit, and what it establishes

1965 → 1985	data	model	
farm emp. share	.59 [†] →.25 [†]	.605→.229	T
mfg VA share	.15→.30	.192→.298	T
real GDP p.c.	×4 [†]	×4.00	T
exports / GDP	.09 [†] →.33 [†]	.078→.330	T
urban share 1980	.57	.547	T
urban share 1985	.65 [†]	.629	U
farm VA share	.38 [†] →.13 [†]	.595→.239	U
OtherAg import share	—	.095→.764	
trad. rice share	~1→<1	1.00→.515	

T targeted, U untargeted. Guess-invariant to 5×10^{-12} .

- Farm employment falls 38 points through **Engel effects** on rising income, **grain import competition**, and **migration**, while the economy grows fourfold and urbanises from 0.32 to 0.63.
- Exports 8% → 33% of GDP requires foreign demand growing 24%/yr at $\sigma = 4$.
- Rice mechanisation takes off only after 1975, as wages rise: the induced-innovation timing, but its level is set by an assumed fixed cost.

Mechanical limits of the current baseline

1. **No agricultural productivity gap.** Rural wage 0.76 of urban in 1965; farm VA share 0.595 vs 0.38[†]. Matching it with one wage per region would need a rural/urban wage ratio of $\sim 1/3$. $\Rightarrow$ occupation block (next slide), not yet recalibrated.
2. **1965 manufacturing share too high** (0.19 vs 0.15) with the final-demand weight at its lower bound: the stylised IO coefficients over-produce manufacturing. $\Rightarrow$ BoK input-output tables.
3. **No relative-price channel** in demand: farm TFP releases labour only through income and import competition, not Baumol.

HCI under this baseline: unresolved, not a finding

The (untuned) HCI package *lowers* the 1985 national manufacturing share (VA -0.013 , employment -0.026) and raises services. The sign comes from the baseline, not one channel: income gains go to non-traded services, and manufacturing has almost no domestic final demand (limit 2). The directional test is a strict expected failure until the IO structure comes from data.

Occupation choice in the Korea baseline (illustration, not recalibrated)

case	1965			1985		
	farm emp.	w_A/w_N	APG	farm emp.	w_A/w_N	APG
integrated ($\varepsilon = \infty$)	0.605	0.827	0.958	0.229	0.868	1.053
$\varepsilon = 10, b_A = 1$	0.561	0.951	1.100	0.219	0.905	1.103
$\varepsilon = 5, b_A = 1$	0.530	1.051	1.214	0.218	0.908	1.107
$\varepsilon = 5, b_A = 1.5$	0.627	0.802	0.930	0.314	0.676	0.803
$\varepsilon = 5, b_A = 2$	0.680	0.653	0.758	0.399	0.545	0.638

1965 amenities re-inverted per case; all other parameters at the integrated calibration. Wage ratios national, employment-weighted. APG = farm VA per worker / non-farm.

1. $b_A = 1$ makes things *worse*: a large farm share then needs a higher farm wage.
2. With a farm wedge the gap **appears and widens** during the transformation ($b_A = 2$: APG 0.76 $\rightarrow$ 0.64). The integrated model cannot do this.
3. The wedge also **slows** the transformation (farm emp. 0.40 vs 0.23 in 1985). Recalibration trades (b_A, ε) off against the Engel parameters; separating them needs farm vs non-farm earnings data.

What the township panel does inside the model

Variable	Model object	Role
Population; status	$L_{i,t}$ (amenities); non-employment share	Inversion, estimation
Employment by sector	$\pi_{k n}; A_{ik}, b_{nk}$	I, E, V (Fig. 8 gradient)
Land (paddy/dry); farm-size classes	land endowments; $F_i(\ell)$	I (mechanisation threshold)
Reservoirs (capacity, date)	irrigation shock to paddy quality	Shock ; event studies
Crops, livestock, stores	crop mix, yields, commercialisation	E (von Thünen), V
Municipal finance	local public investment by function; transfers	Shock
Roads paved/unpaved, bridges	local access $a_{i,t}$; annual paving dates	Shock
Electricity; schools	non-farm productivity; cohort human capital	Shock, I, E

I = inversion input; E = estimation moment; V = held-out validation.

Identification map: θ, φ from price gaps (Donaldson) or indirect inference on market-access event studies; ν, δ from O-D gravity and interchange openings; ε, b from shift-share industrial pull; ρ_k from park SDID and the post-1979 wind-down; η, v_j from household-survey Engel curves; mechanisation from adoption by farm-size class $\times$ wage and machine-price shocks.

The reservoir design: the empirical side of Matsuyama in space

- Event studies of reservoir completions, heterogeneous by **predetermined** access to *crop markets* and, **separately**, to *non-farm employment*.
- Treatment: irrigation *command area* and water delivery, not the township containing the dam. Handle construction jobs, roads, displacement, downstream effects.
- Placebo: dry-field townships.
- Use the farm employment *share* (migration contaminates levels).
- A positive interaction is not a reversal: show the irrigation effect **crosses zero** within the observed support.

Channel	Evidence
Commercialisation	marketed output, farm-gate prices, farm labour demand, land returns
Labour release	labour per hectare ↓, mechanisation, non-farm work, migration
Local demand spillovers	rural services employment after farm-income gains

Model side: derive the predicted coefficients with θ, φ, η set elsewhere, then compare (indirect inference).

Build order

Stage	Content	Deliverable
0 (done)	Closure, measurement, nested choice, σ_x , equivalent income; history baseline; occupation block	Numbers that measure what their labels say
1	Small spatial agriculture model: paddy/dry, competitive farms, factor-biased irrigation, non-farm sector, land, trade, mobility	Commercialisation-vs-release prediction
2	Reservoir event studies (command areas, separate access measures)	Empirical side, standalone
3	Annual clock, gravity, inversion (after identification), occupation with non-employment, structural event studies	Where and when fundamentals moved
4	Capital, schooling, demography, poverty; Shapley decomposition	The macro paper

Stages 1–2 come first because they test the mechanism behind “Matsuyama in space” before a large model exists. Annual timing, competitive agriculture and a simpler trade block are sensible from Stage 1.

To agree on (recommendations from *fresh_look.md* §9)

1. Adopt “baseline = history + decomposition” with poverty and structural transformation as headline outcomes? (recommended)
2. Drop Melitz in the macro model; keep firm heterogeneity for an MMS satellite? (recommended)
3. Capital: observed aggregate investment first, endogenous saving later? (recommended)
4. Geography: townships + urban units, counties for manufacturing? Needs urban coverage confirmed; consider commuting zones (township data are by residence).
5. Closure data: feed the observed NX/GDP path once the model is annual? (recommended)

And the near-term modelling calls

- Recalibrate with the occupation block on, once farm/non-farm earnings are in.
- Replace stylised IO coefficients with BoK tables before reading any HCI number.
- Calibrate σ_x externally; report headlines with its sensitivity.

Data we most need (cheap now, expensive after the freeze)

To recalibrate the current model

- Farm vs non-farm earnings (APG; separates b_A , ϵ , Engel)
- Food expenditure share by income (PIGL directly)
- Grain self-sufficiency / import shares
- Rice's share of food spending
- Power tillers per farm household, 1965–85
- Sectoral productivity growth (GGDC 10-sector)
- BoK input-output tables
- Regional population and farm employment
- Verify every $\dagger$ target against source

Asks for the digitisation team

- Urban units (dong/gu/si): the destination side of every flow
- Annual boundary crosswalk with area weights
- Variable-year metadata; double-entry sample
- Local prices: rice, barley, fertiliser, tillers, farm wages
- Irrigated vs rain-fed paddy; output by crop and **rice variety** (Tongil); machinery stocks; landholding classes in full
- Population by age and sex; births and deaths
- Municipal finance chart of accounts; schools entrants/graduates

Farm productivity growth: commercialisation or labour release?

It depends on where you are.
Korea lets us measure where.

Backup: how the old HCI headline moved

1985, policy — baseline (old 3-sector economy)	old closure	determinate	+ nested	+ ag recal.
heavy-mfg gross-output share	+0.057	+0.104	+0.095	+0.103
agriculture share	−0.041	−0.045	−0.040	−0.005
services share	−0.016	−0.060	−0.055	−0.098
Changwon wage	+0.144	+0.120	+0.122	+0.087
rural income per capita	+0.199	+0.171	+0.174	+0.121

- Pre-accounting-fix code (*cdafe29*) gave +0.127 and was determinate. Old accounts → corrected accounts with balanced trade: +0.127 → +0.104. The further fall to +0.057 was the indeterminate closure's selection.
- All of this is superseded by the history-calibrated 4-sector baseline, under which the HCI aggregate effect is unresolved.

Backup: counterfactuals that tell the story

1. **Decomposition of 1962–90 growth and poverty reduction** over drivers $D = \{\text{world demand, capital, human capital, demography, HCI credit, parks, roads, rural programmes, residual TFP}\}$:

$$\phi_d = \sum_{S \subseteq D \setminus \{d\}} \frac{|S|! (|D| - |S| - 1)!}{|D|!} [Y(S \cup \{d\}) - Y(S)].$$

10 drivers = 1,024 dynamic solves; 5 blocks = 32.

2. **Complementarity**: $Y(a, b) - Y(a) - Y(b) + Y(\emptyset)$ for parks $\times$ roads, roads $\times$ rural programmes, schooling $\times$ HCI.
3. **Sequencing**: roads before parks; Green Revolution before mechanisation.
4. **Incidence**: who escaped poverty, by origin-township type and land class.
5. **Persistence / big push**: remove HCI wedges after 1979.

Poverty headcount from cell incomes with lognormal dispersion: $H_t = \sum_c \omega_{c,t} \Phi\left(\frac{\ln z - \ln \bar{y}_{c,t}}{\sigma_c} + \frac{\sigma_c}{2}\right)$.

Backup: derivation of the township equation

Competitive farm, Cobb–Douglas: $p(1 - \alpha)Q/L_A = w$ with $Q = (BT)^\alpha L_A^{1-\alpha}$ gives

$$\log L_A = \frac{1}{\alpha} [\log(1 - \alpha) + \log p + \alpha \log(BT) - \log w].$$

Faced demand $Q = D p^{-e}$, so $d \log p = -\frac{1}{e} d \log Q = -\frac{1}{e} [\alpha d \log B + (1 - \alpha) d \log L_A]$. Substituting and collecting $d \log L_A$:

$$d \log L_A \left(\alpha + \frac{1 - \alpha}{e} \right) = \alpha \left(1 - \frac{1}{e} \right) d \log B - d \log w \implies d \log L_A = \frac{\alpha(e - 1) d \log B - e d \log w}{\alpha e + 1 - \alpha}.$$

- $e \rightarrow \infty$ (price-taker): $d \log L_A = d \log B - \frac{1}{\alpha} d \log w$, the general form in the referee report.
- Note the release term is *larger* where e is large: connected townships respond more to the opportunity wage as well.
- Partial equilibrium and Hicks-neutral; the quantitative model replaces both.

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Internal: *docs/fresh_look.md*, *docs/review_fresh_look.md*, *docs/model.tex*, *scripts/data_targets.py*.